11A
MOLECULAR SHAPE

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

CHAPTER 11

1. Use VSEPR theory and Lewis structures to predict molecular shapes.
2. Identify polar and nonpolar molecules using electronegativity values and molecular shape.
3. Explain valence bond theory.
4. Describe and recognize *sp*, sp^2, and sp^3 hybridization based on the Lewis structure of a molecule.
5. Define, describe, and identify sigma (σ-) and pi (π-) bonds in molecules.

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?

Chemical Formulas → Atomic Structure → Chemical Bonding → Molecular Structure → Molecular Properties

covalent compound → chemical formula → Lewis structure → VSEPR theory

chemical formula:
- composition

Lewis structure:
- 2D structure
- bond polarity

VSEPR theory:
- 3D shape
- molecular polarity

VSEPR and Molecular Geometry

11.1

Valence shell electron pair repulsion (VSEPR) theory

- Used to predict the shape of a molecule based on minimizing repulsion between electron domains around the central atom.
- How to identify and count electron domains?

each single bond	–	one domain
each double bond	=	one domain
each triple bond	≡	one domain
each lone pair	··	one domain

misconceptions

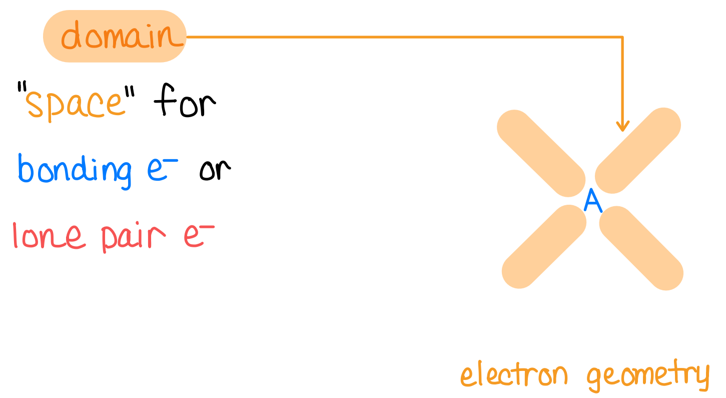
- a double bond is not two
- a triple bond is not three

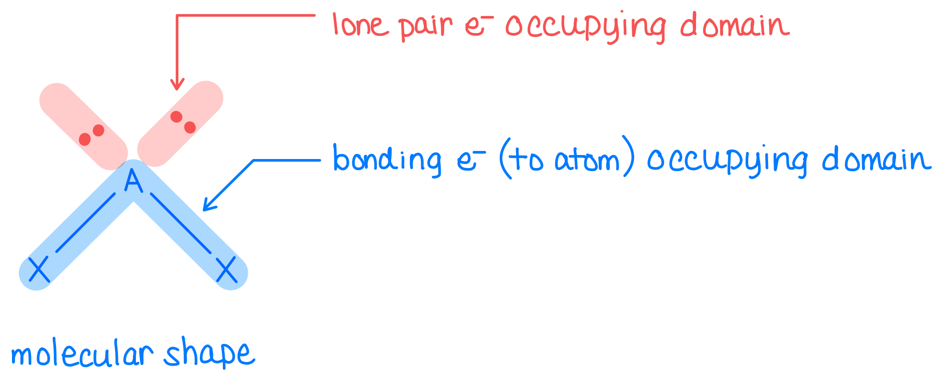
VSEPR and Molecular Geometry

11.1

Electron geometry looks at the arrangement of total *electron domains*.

- Most stable arrangement of electron domains around the central atom (A) because it results in the *least repulsion* between domains.





Molecular shape looks only at how the *atoms* are arranged in the molecule.

look only at blue highlights!

VSEPR and Molecular Geometry

11.1

Total number of domains	Electron geometry	Molecular shape
2 domains	linear	?
3 domains	trigonal planar	?
4 domains	tetrahedral	?
5 domains	trigonal bipyramidal	?
6 domains	octahedral	?

There are five electron geometries.

- Based on number of electron domains on the central atom.
- If no lone pairs on central atom:

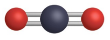
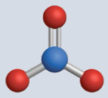
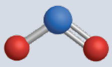
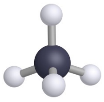
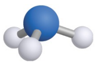
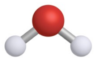
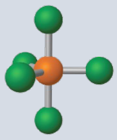
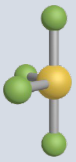
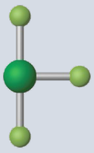
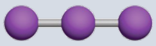
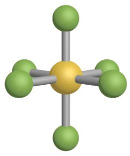
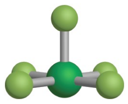
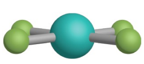
  electron geometry = molecular shape

- If lone pairs on central atom:

  electron geometry ≠ molecular shape

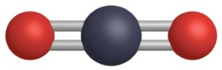
VSEPR and Molecular Geometry

11.1

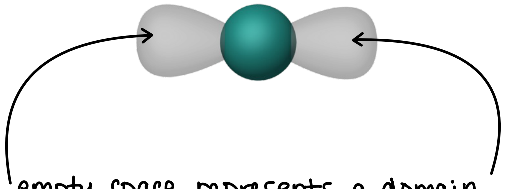
Domains	Lone pairs	Bond domains	Electron geometry	Molecular shape	Bond Angles	Example + Model
2	0	2	linear	linear	180°	CO_2 
3	0	3	trigonal planar	trigonal planar	120°	NO_3^- 
3	1	2	trigonal planar	bent	<120°	NO_2^- 
4	0	4	tetrahedral	tetrahedral	109.5°	CH_4 
4	1	3	tetrahedral	trigonal pyramidal	<109.5°	NH_3 
4	2	2	tetrahedral	bent	<109.5°	H_2O 
5	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 
5	1	4	trigonal bipyramidal	seesaw	<90° <120° <180°	SF_4 
5	2	3	trigonal bipyramidal	t-shaped	<90° <180°	ClF_3 
5	3	2	trigonal bipyramidal	linear	180°	I_3^- 
6	0	6	octahedral	octahedral	90° 180°	SF_6 
6	1	5	octahedral	square pyramidal	<90° <180°	ClF_5 
6	2	4	octahedral	square planar	90° 180°	XeF_4 

VSEPR and Molecular Geometry

11.1

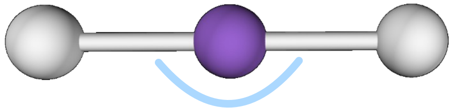
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
2 domains	0	2	linear	linear	180°	CO_2 

Electron geometry



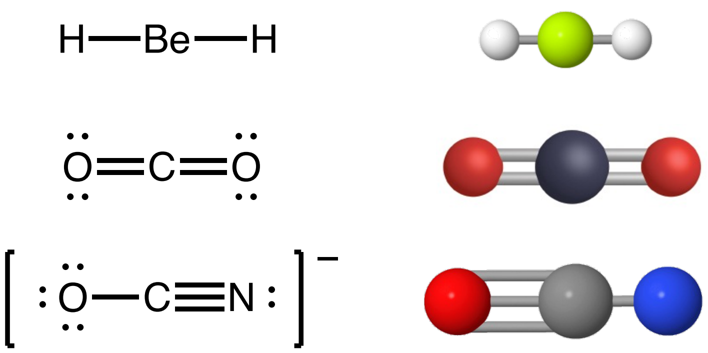
empty space represents a domain; can be occupied by bond to atoms or lone pairs

Molecular shape



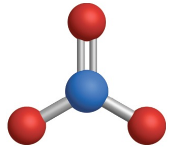
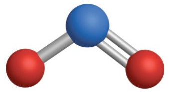
180° bond angle

Other examples

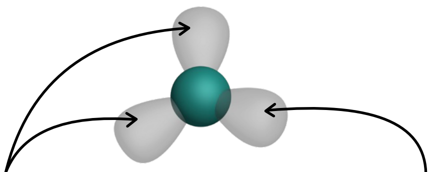


VSEPR and Molecular Geometry

11.1

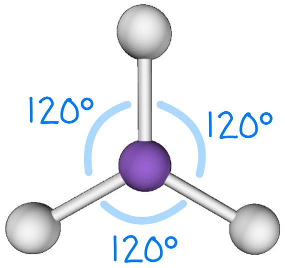
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
3 domains	0	3	trigonal planar	trigonal planar	120°	NO_3^- 
3 domains	1	2	trigonal planar	bent	< 120°	NO_2^- 

Electron geometry

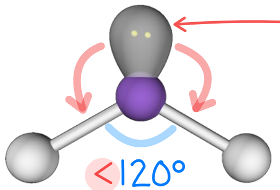


empty space represents a domain;
can be occupied by
bond to atoms or lone pairs

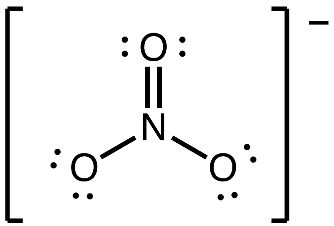
Molecular shapes

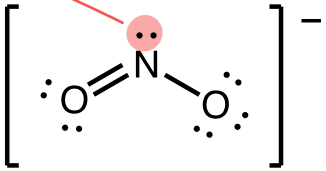


lone pairs repel bonded atoms
so bond angle < 120°



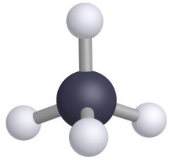
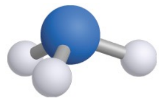
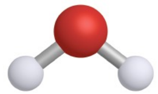
Lewis structures



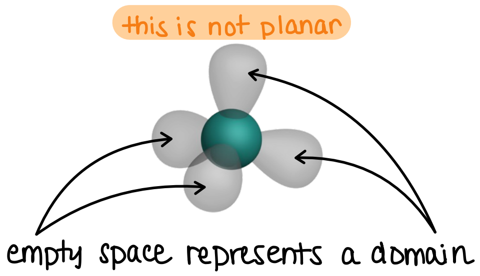


VSEPR and Molecular Geometry

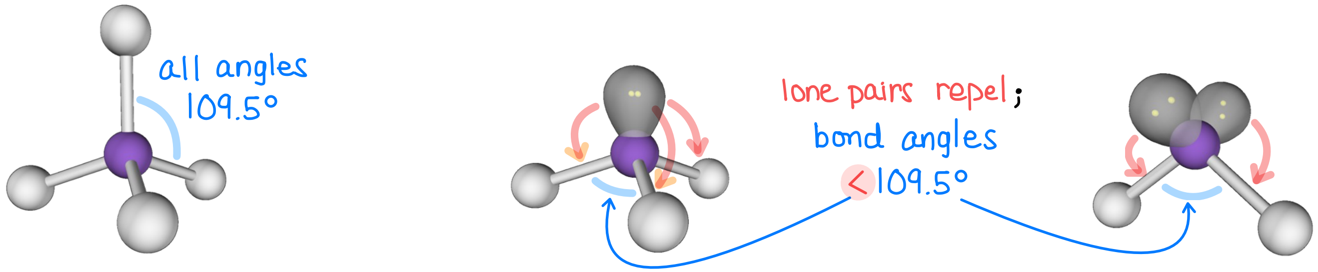
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model	
4 domains	0	4	tetrahedral	tetrahedral	109.5°	CH_4	
4 domains	1	3	tetrahedral	trigonal pyramidal	< 109.5°	NH_3	
4 domains	2	2	tetrahedral	bent	< 109.5°	H_2O	

Electron geometry



Molecular shapes



VSEPR and Molecular Geometry

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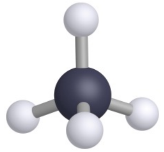
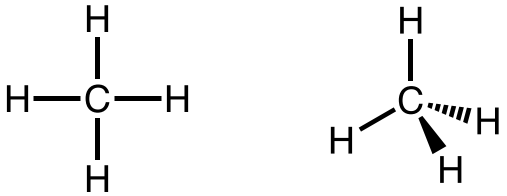
4 domains	0	4	tetrahedral	tetrahedral	109.5°	CH_4 

2D Lewis structure of methane



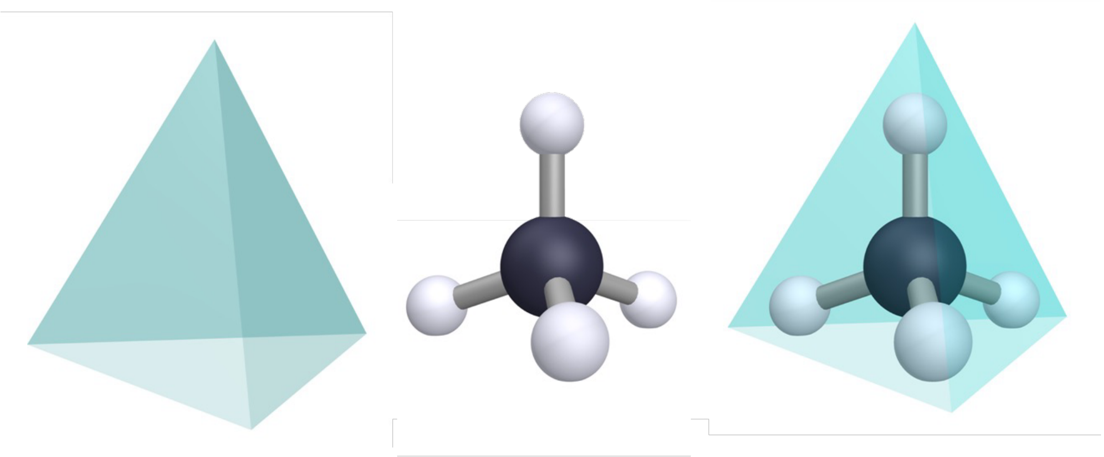
chemist's note:

bond in plane of paper

bond coming out of plane of paper

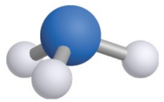
bond going into plane of paper

3D structure of methane

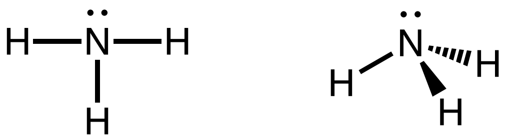


VSEPR and Molecular Geometry

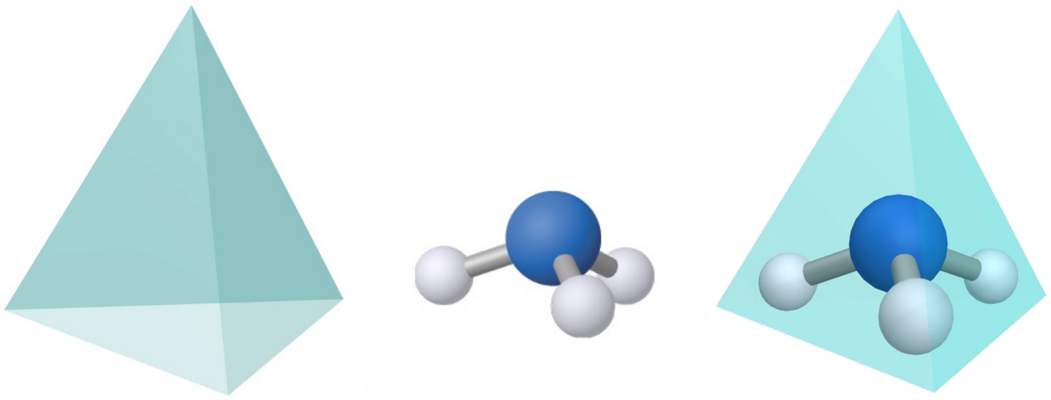
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
4 domains	1	3	tetrahedral	trigonal pyramidal	< 109.5°	NH_3 

2D Lewis structure of ammonia

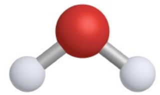


3D structure of ammonia



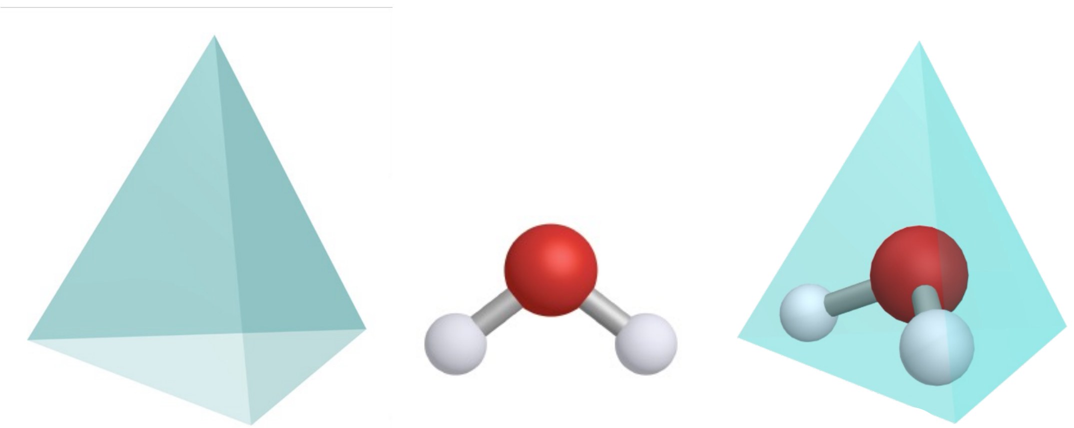
VSEPR and Molecular Geometry

11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
4 domains	2	2	tetrahedral	bent	< 109.5°	H_2O 

2D Lewis structure of water

3D structure of water



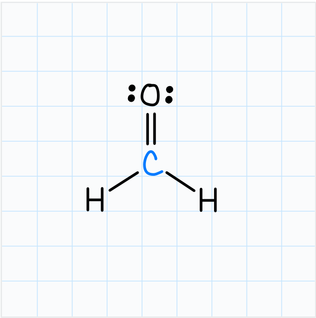
IN-CLASS QUESTION 1

Learning Objective(s): 11.1

Draw the Lewis structure for formaldehyde, H_2CO.

less electronegative atom in center

$2(1e^-) + 4e^- + 6e^- = 12e^-$



Then, complete the table for the molecule to describe the molecular shape.

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape
3	0	3	trigonal planar	trigonal planar

VSEPR and Molecular Geometry

11.1

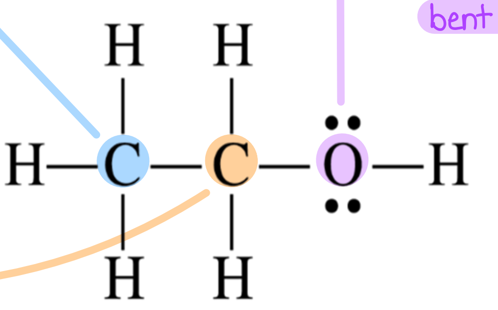
Molecules with multiple central atoms

- Any “non-terminal” atom (an atom with only one bonding domain) is considered a “central” atom.
- Shapes of larger molecules are usually described as a series of smaller shapes linked together.

4 electron domains,
0 lone pairs,
tetrahedral electron geometry,
tetrahedral molecular shape

4 electron domains,
0 lone pairs,
tetrahedral electron geometry,
tetrahedral molecular shape

4 electron domains,
2 lone pairs,
tetrahedral electron geometry,
bent molecular shape



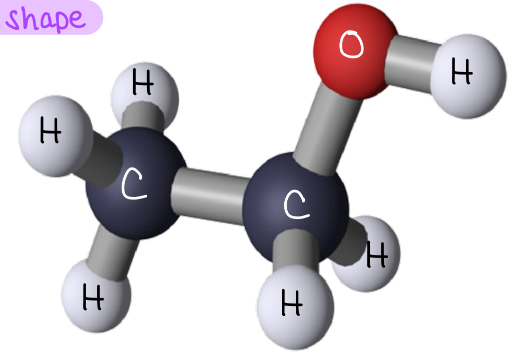
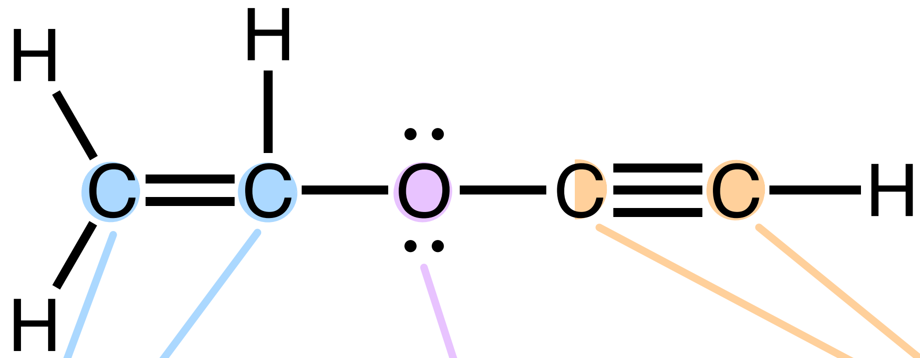


FIGURE 11.10 Lewis structure and 3D model of ethanol (CH_3CH_2OH)

IN-CLASS QUESTION 2

Learning Objective(s): 11.1

Determine the molecular shape for each central atom in the molecule drawn below.



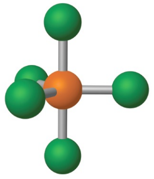
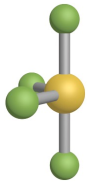
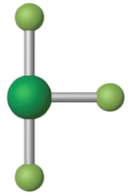
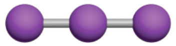
3 electron domains,
0 lone pairs,
trigonal planar electron geometry,
trigonal planar molecular shape

4 electron domains,
2 lone pairs,
tetrahedral electron geometry,
bent molecular shape

2 electron domains,
0 lone pairs,
linear electron geometry,
linear molecular shape

VSEPR and Molecular Geometry

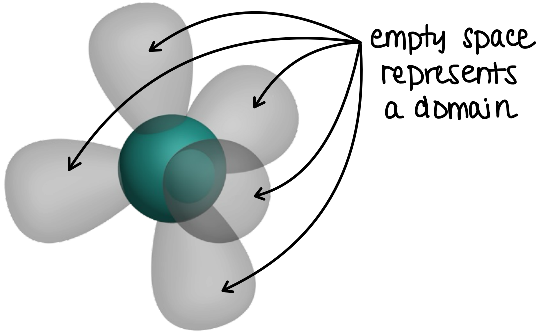
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 
5 domains	1	4	trigonal bipyramidal	seesaw	< 90° < 120° < 180°	SF_4 
5 domains	2	3	trigonal bipyramidal	T-shaped	< 90° < 180°	ClF_3 
5 domains	3	2	trigonal bipyramidal	linear	180°	I_3^- 

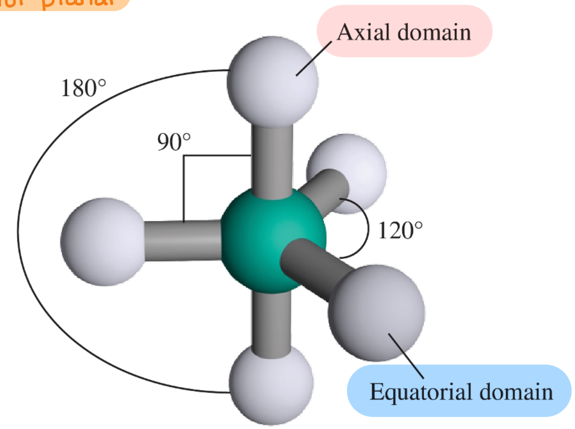
VSEPR and Molecular Geometry

11.1

Electron geometry



this is not planar

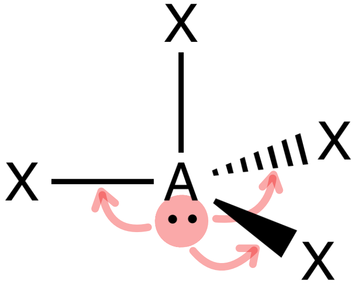


For trigonal bipyramid electron geometry:

Lone pairs occupy equatorial positions!

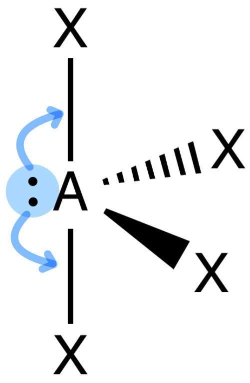
Axial ×

lots of repulsion



Equatorial ✓

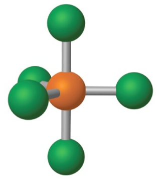
less repulsion



Axial	Equatorial
Three 90° e^- pair repulsions	Two 90° e^- pair repulsions
One 180° e^- pair repulsion	Two 120° e^- pair repulsions

VSEPR and Molecular Geometry

11.1

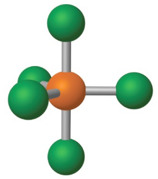
Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 

Molecular shapes

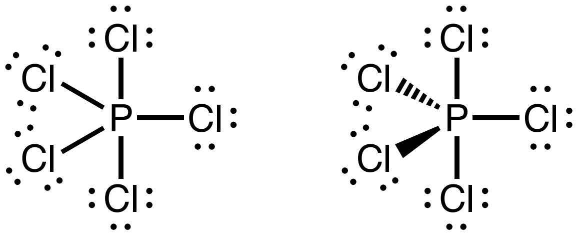
lone pairs only occupy equatorial domains!

VSEPR and Molecular Geometry

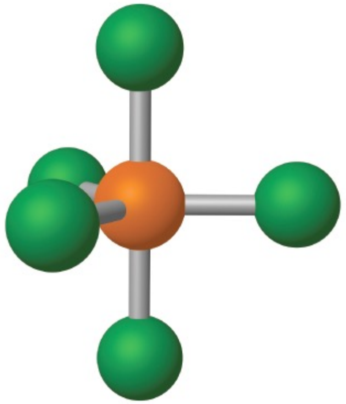
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	0	5	trigonal bipyramidal	trigonal bipyramidal	90° 120° 180°	PCl_5 

2D Lewis structure

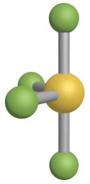


3D structure

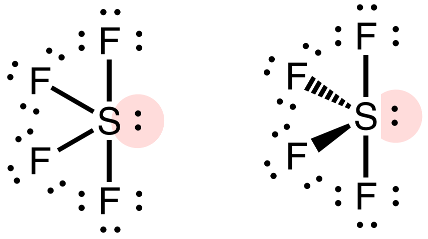


VSEPR and Molecular Geometry

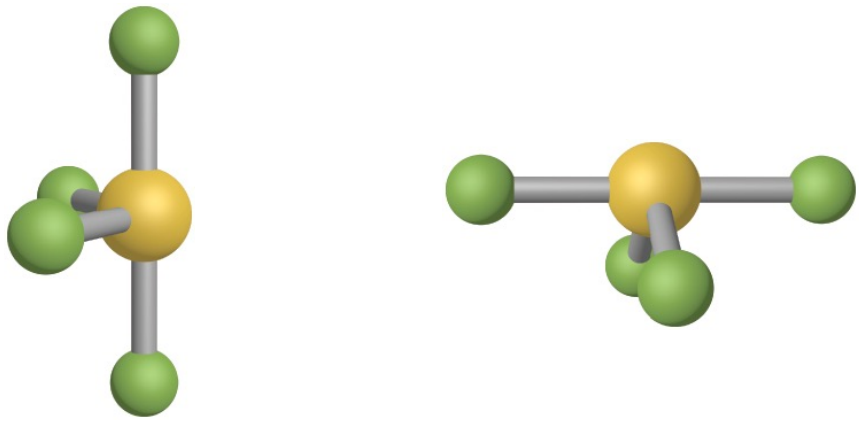
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	1	4	trigonal bipyramidal	seesaw	< 90° < 120° < 180°	SF_4 

2D Lewis structure

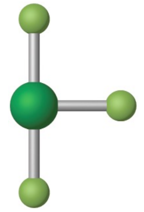


3D structure

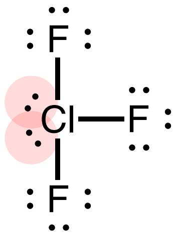


VSEPR and Molecular Geometry

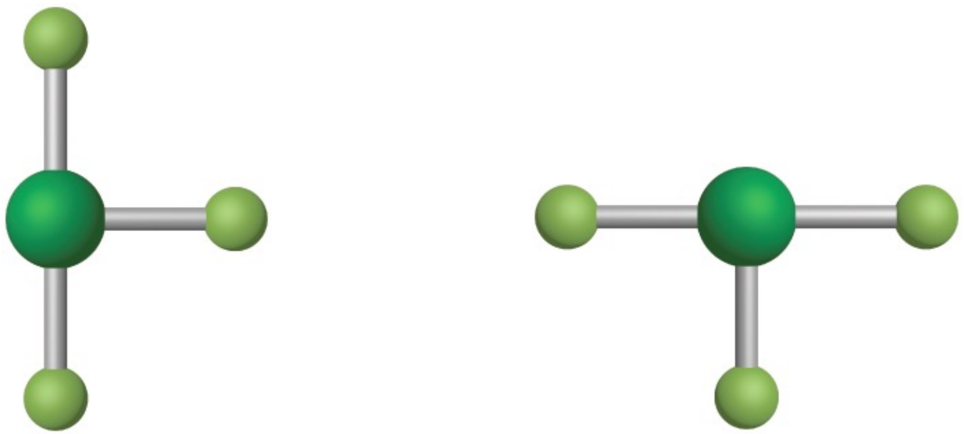
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	2	3	trigonal bipyramidal	T-shaped	< 90° < 180°	ClF_3 

2D Lewis structure

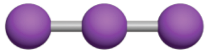


3D structure

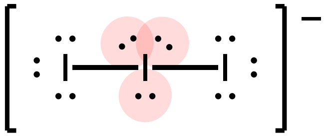


VSEPR and Molecular Geometry

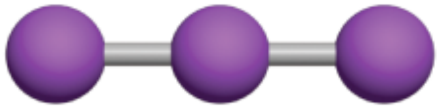
11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
5 domains	3	2	trigonal bipyramidal	linear	180°	I_3^- 

2D Lewis structure

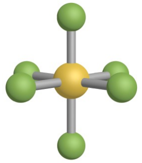
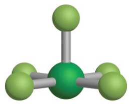
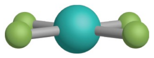


3D structure



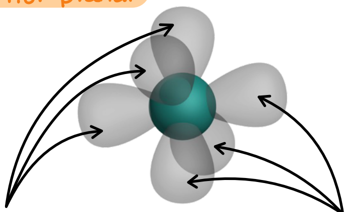
VSEPR and Molecular Geometry

11.1

Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape	Bond Angles	Example + Model
6 domains	0	6	octahedral	octahedral	90° 180°	SF_6 
6 domains	1	5	octahedral	square pyramidal	< 90° < 180°	ClF_5 
6 domains	2	4	octahedral	square planar	90° 180°	XeF_4 

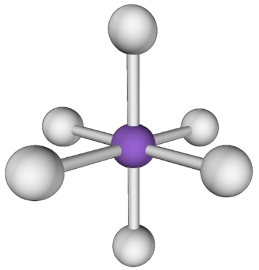
Electron geometry

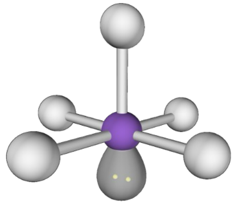
this is not planar



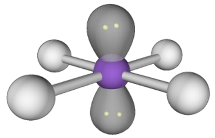
empty space represents a domain

Molecular shapes



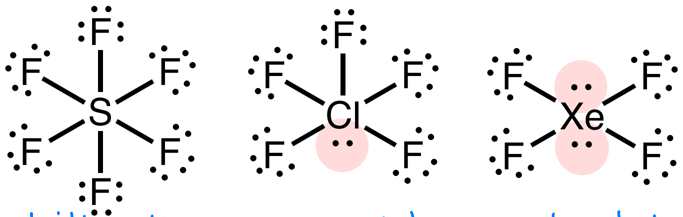


one lone pair



two lone pairs

Lewis structures



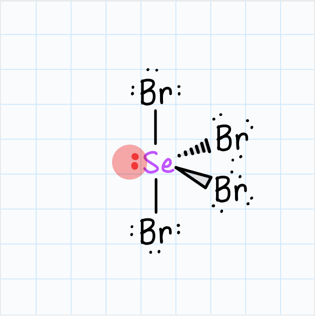
counterintuitive: there are no axial or equatorial domains

EXAMPLE PROBLEM

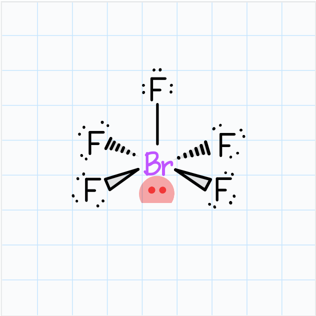
Learning Objective(s): 11.1

Draw the Lewis structures for $SeBr_4$ and BrF_5.

$6e^- + 4(7e^-) = 34e^-$



$7e^- + 5(7e^-) = 42e^-$



chemist's note:
- bond in plane of paper
- bond coming out of plane of paper
- bond going into plane of paper

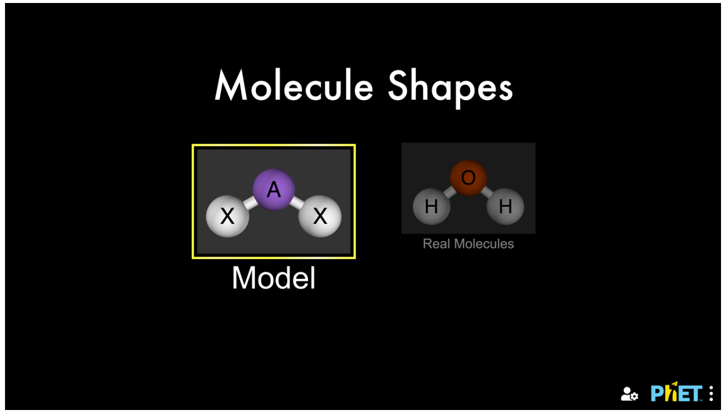
Then, complete the table for the molecules to describe their molecular shapes.

Molecule	Total Number of Domains	# of Lone Pairs	# of Bond Domains	Electron Geometry	Molecular Shape
$SeBr_4$	5	1	4	trigonal bipyramidal	seesaw
BrF_5	6	1	5	octahedral	square pyramidal

Visualizing 3D Molecular Shapes

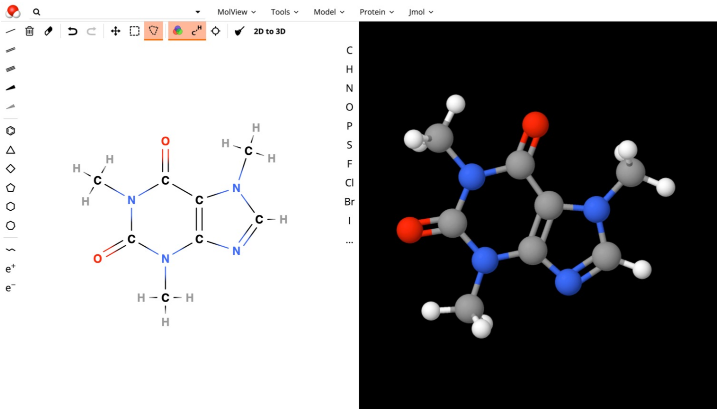
Chemistry in Context

Virtual Tools



https://phet.colorado.edu/sims/html/molecule-shapes/latest/molecule-shapes_all.html

https://molview.org/



Physical Tools

VSEPR model kits

Monday, April 14

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Identify the molecular shape (also called molecular geometry) for the hypothetical molecule XF_3, where X is an element with 7 valence electrons.

- ❑ Linear
- ❑ Trigonal planar
- ❑ Bent
- ❑ Tetrahedral
- ❑ Trigonal pyramidal
- ❑ Trigonal bipyramidal
- ❑ Seesaw
- ❑ T-shaped
- ❑ Octahedral
- ❑ Square pyramidal
- ❑ Square planar